SIMATS ENGINEERING
ASSESSMENT TOOL 2

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COURSE NAME:CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem *(2)*	Application of Engineering Concepts *(2.5)*	Solution Design *(2.5)*	Use of Tools *(2)*	Analysis *(1)*	*Total(10)*
Weightage	*20%*	25%	25%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I __ Siva Dakshitha K /192425395__ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Siva Dakshitha K
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

ANSWER: Supermarket Purchases

Clustering Technique: K-Means Clustering

K-Means is suitable for segmenting customers according to their shopping behavior. The features **Visits/Month, Average Spend, and Items Bought** are used for clustering.

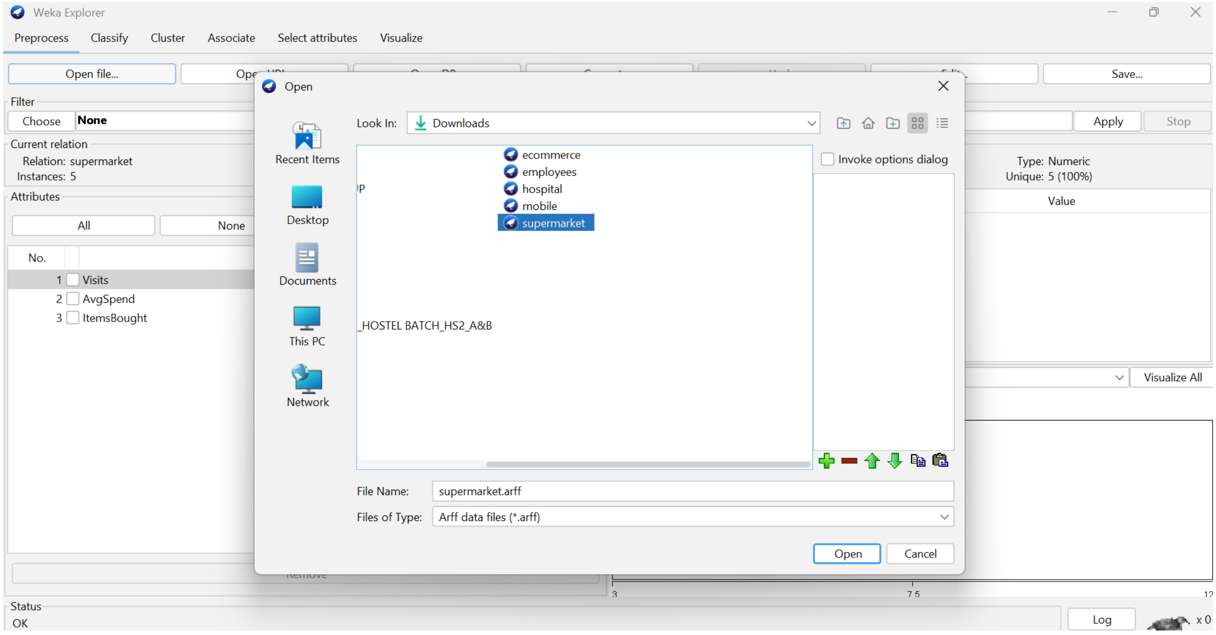
Input

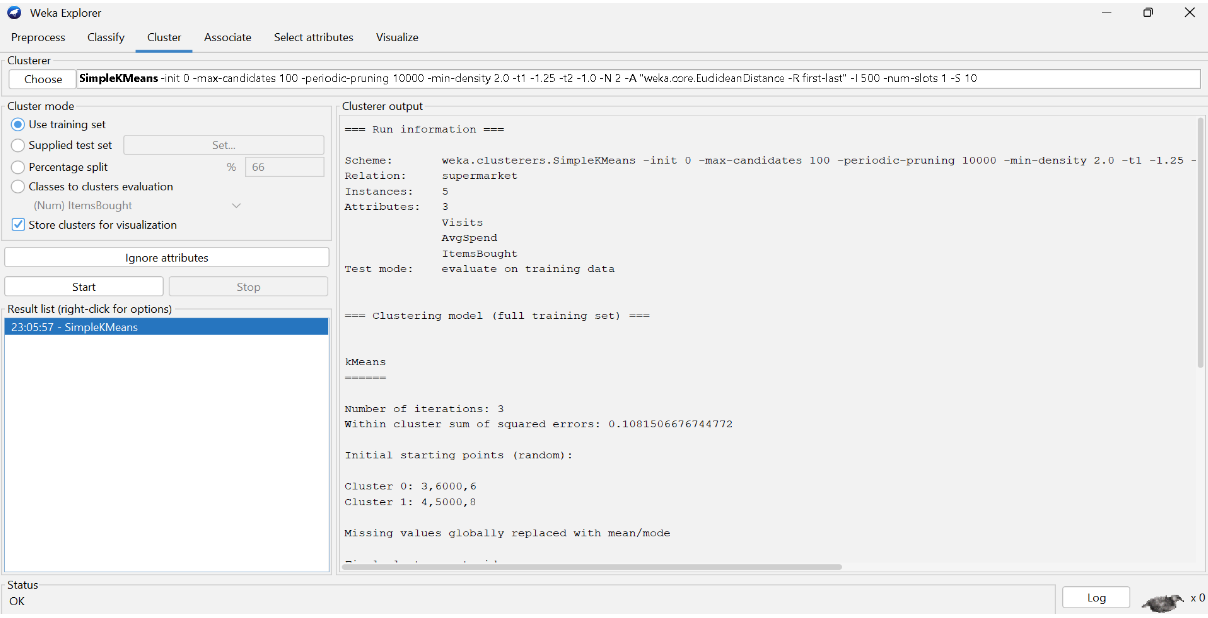
Customer	Visits	Avg Spend	Items
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

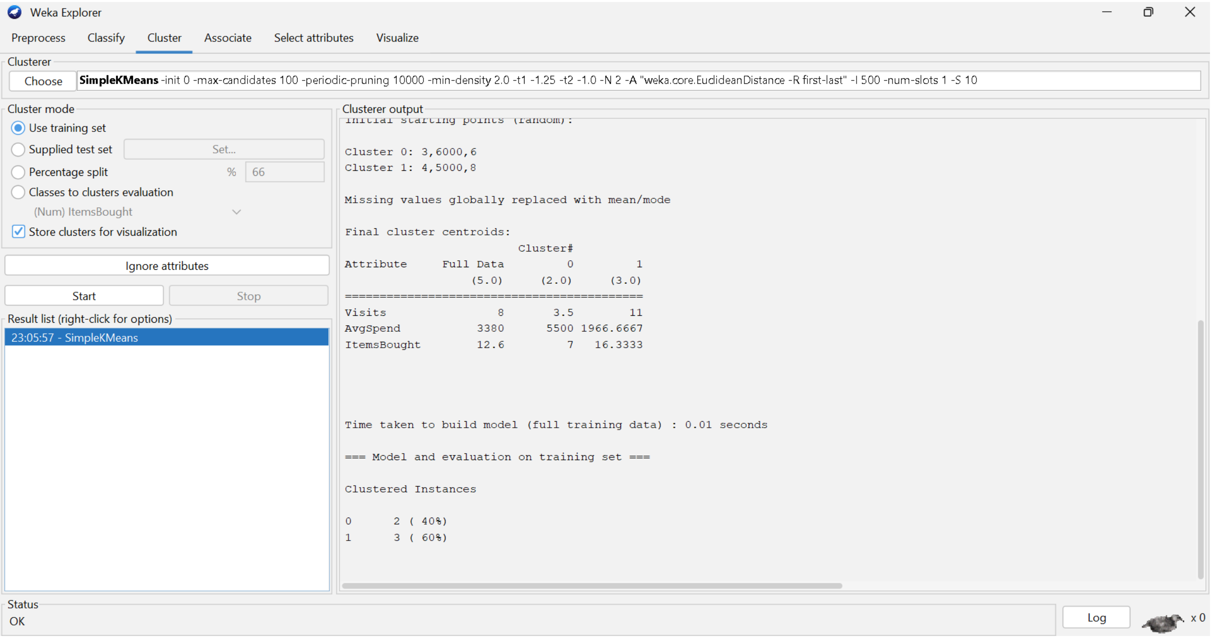
Industry Application

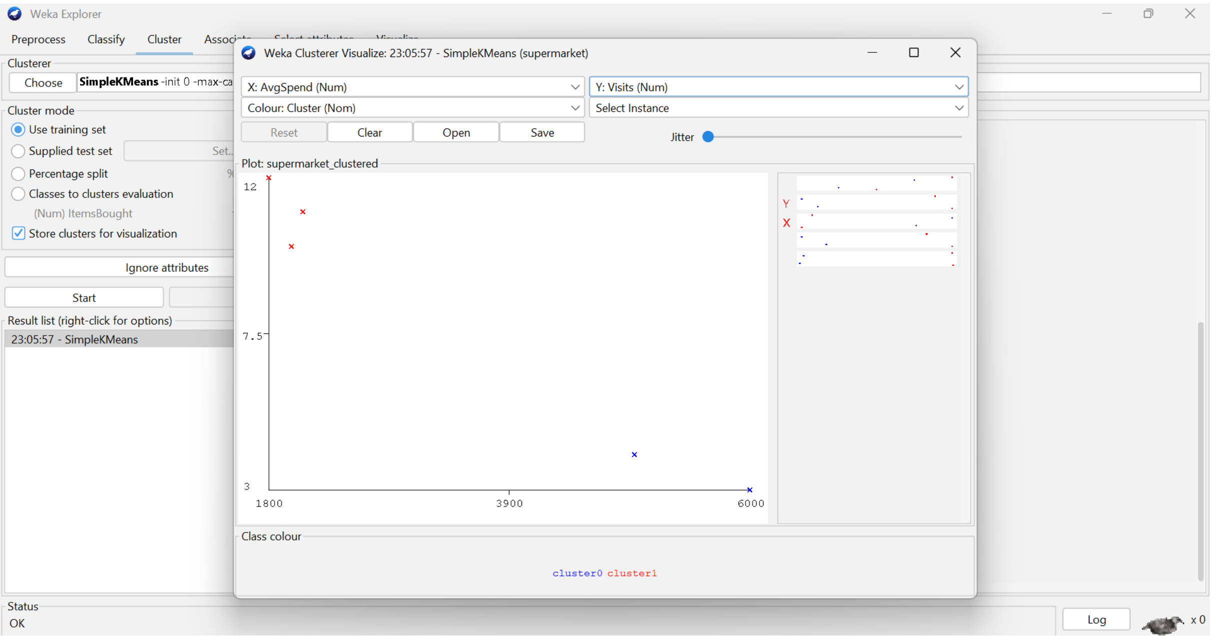
The supermarket can provide loyalty rewards and frequent-purchase offers to Cluster 1, while Cluster 2 can receive premium offers and incentives to increase visit frequency.

OUTPUT:









Analysis

Using K-Means with **K = 2**, customers can be divided into two behavioral groups:

- **Cluster 1:** C1, C3, C5 — frequent visitors with more items purchased.
- **Cluster 2:** C2, C4 — less frequent but higher-spending customers.

Conclusion

K-Means helps the supermarket identify customer segments based on purchasing behavior, enabling personalized and targeted marketing.

Q2. Dataset: Credit Card Transactions

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:
Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

ANSWER: Credit Card Transactions

Clustering Technique: K-Means Clustering

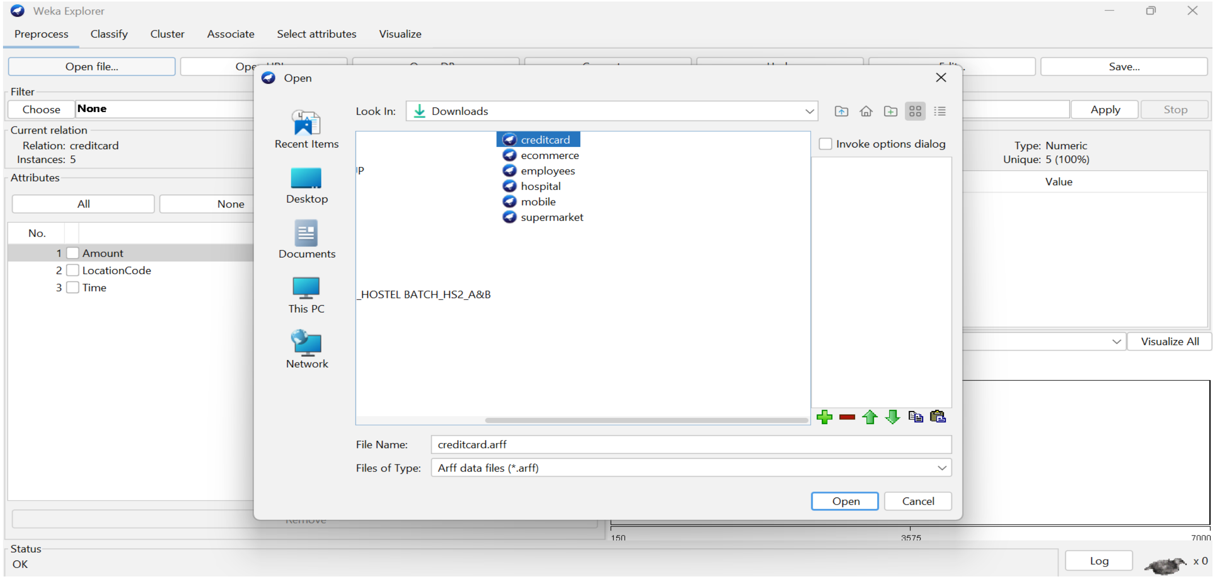
K-Means can group transactions according to transaction amount, location, and transaction time. Unusual transactions that differ significantly from normal clusters can be flagged for investigation.

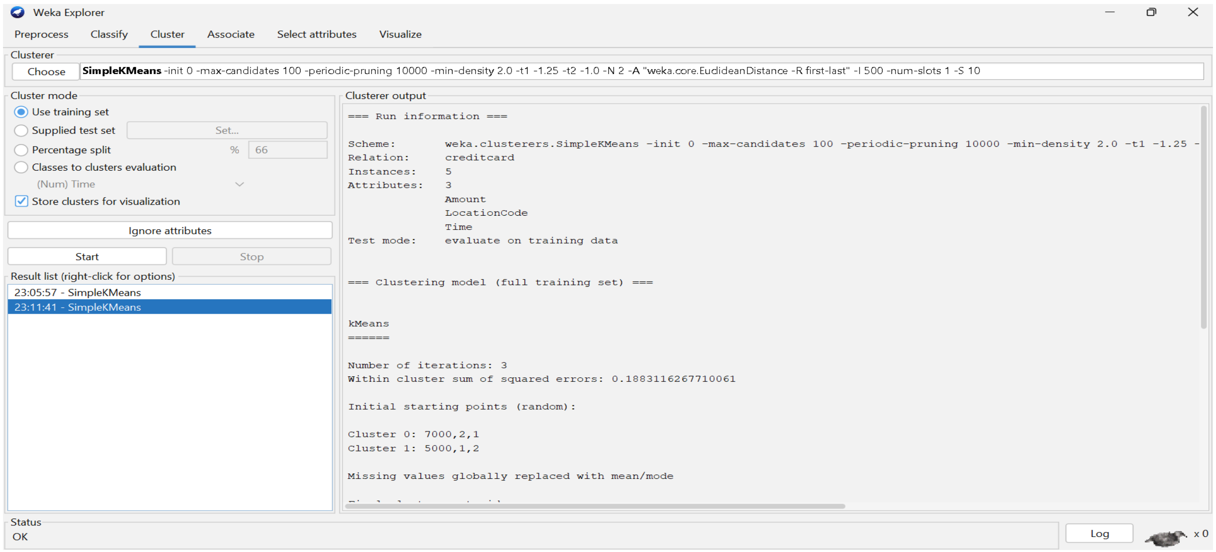
Input

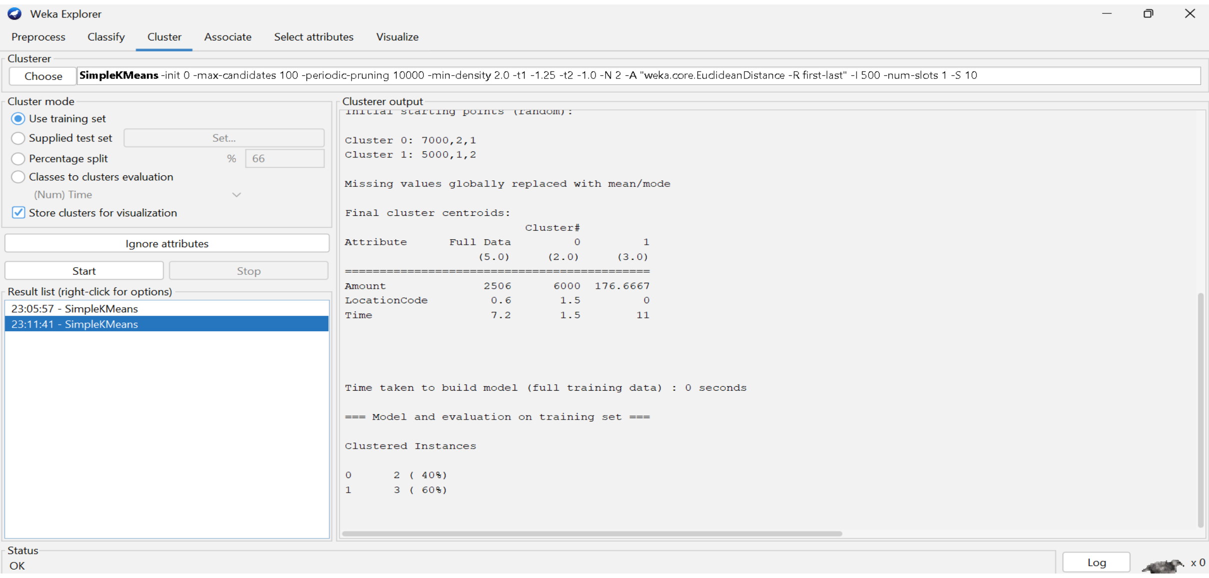
Transaction	Amount	Location	Time
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

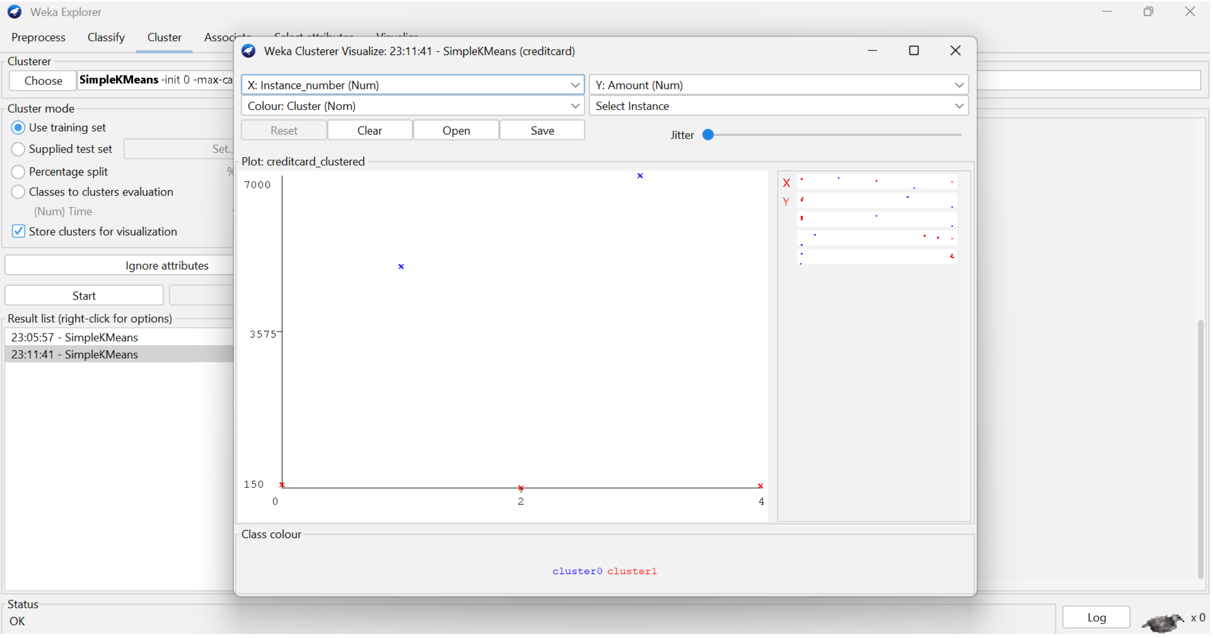
Location is encoded numerically before clustering.

OUTPUT:









Analysis

The transactions T1, T3, and T5 show similar characteristics: low amounts, Chennai location, and daytime transactions.

T2 and T4 have significantly higher transaction amounts and occur during early morning hours in different locations. Therefore, they can be identified as **potential anomalies**.

Industry Application

The bank can flag unusual transactions for additional authentication, customer verification, or fraud investigation.

Conclusion

Clustering can be used as an anomaly-detection technique to identify transactions that differ significantly from normal transaction patterns. It does not by itself prove fraud but helps prioritize transactions for investigation.

Q3. Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

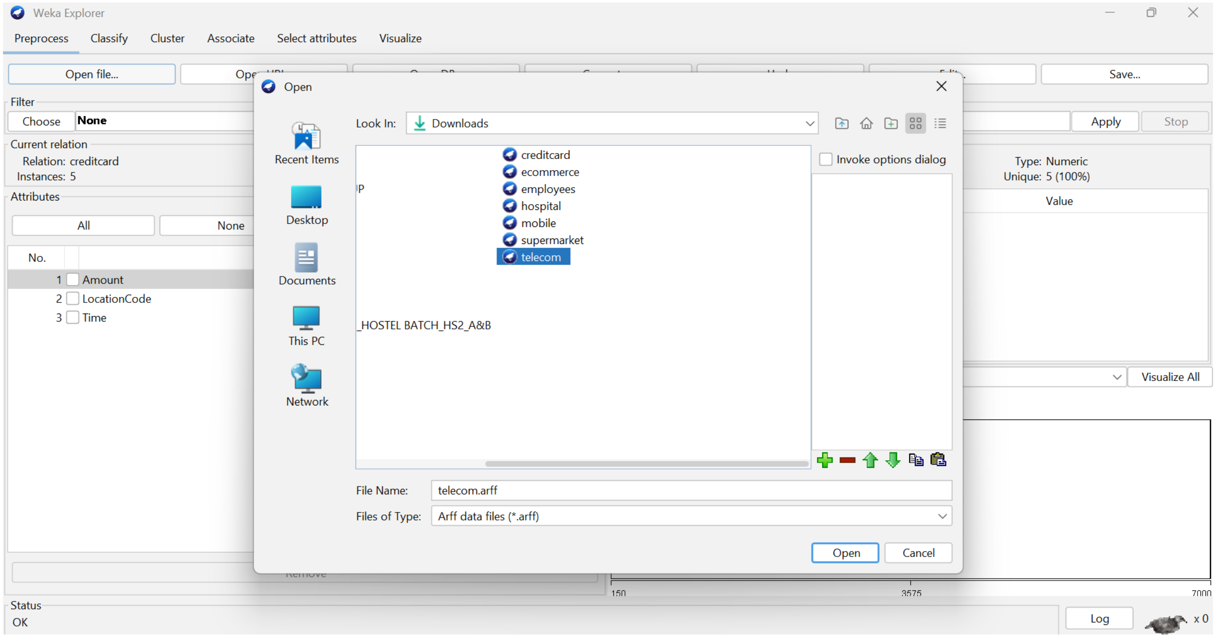
Question:
Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

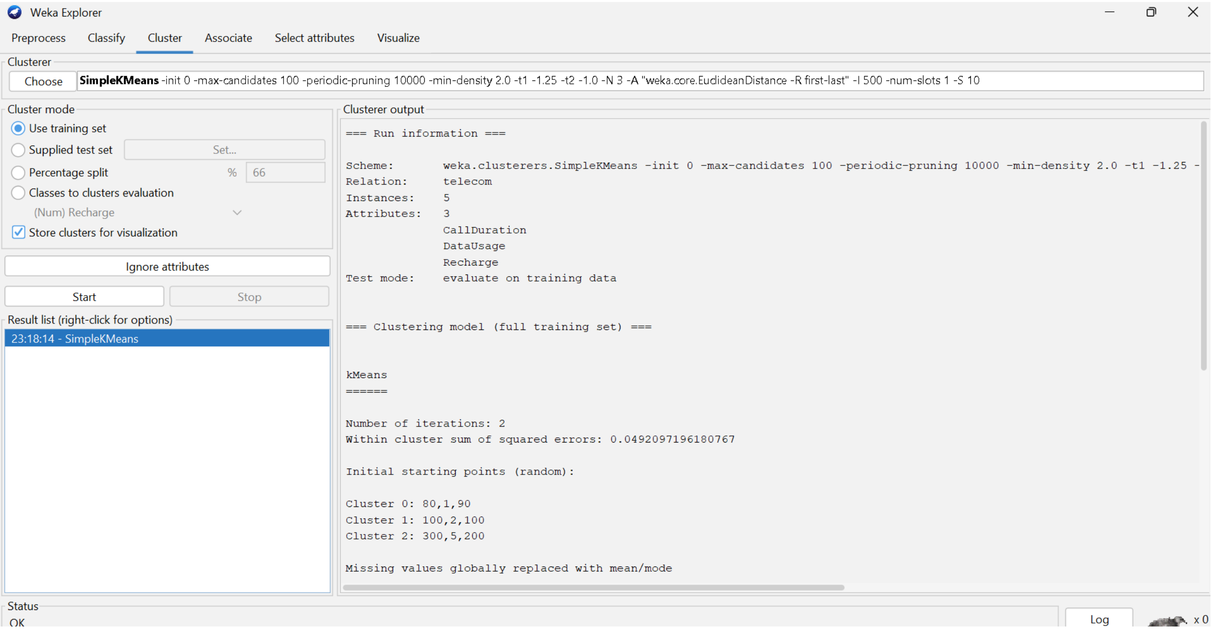
ANSWER: Telecom Usage

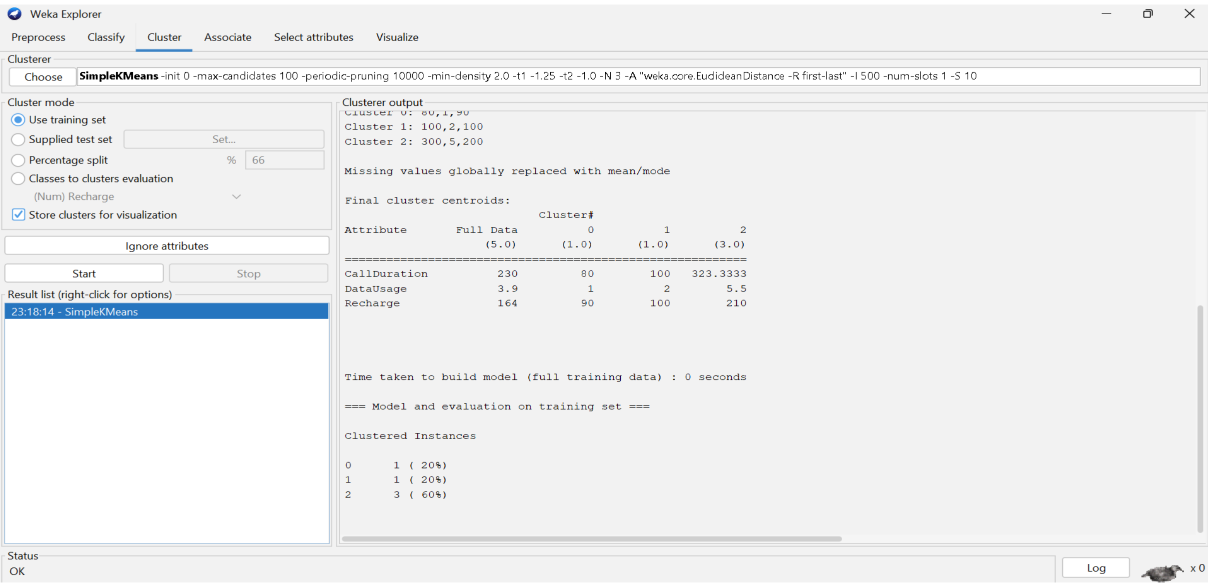
Clustering Technique: K-Means Clustering
K-Means can group telecom users according to **call duration, data usage, and recharge amount**.
Input

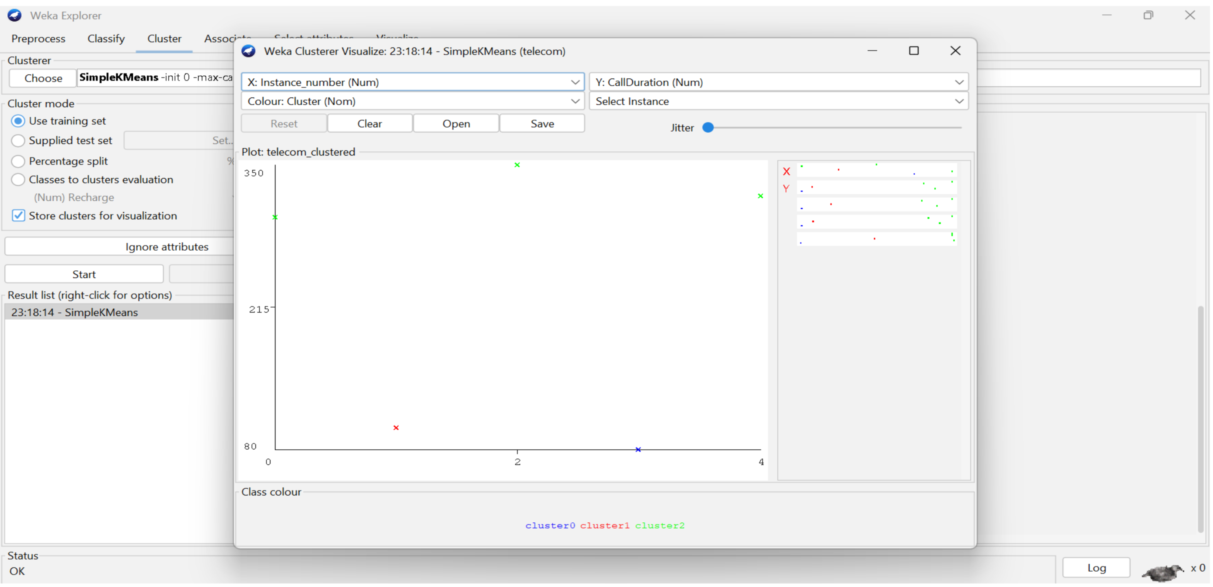
User	Call Duration	Data Usage	Recharge
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

OUTPUT:




Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Recharge
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
23:18:14 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation: telecom
Instances: 5
Attributes: 3
CallDuration
DataUsage
Recharge
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 2
Within cluster sum of squared errors: 0.0492097196180767
Initial starting points (random):
Cluster 0: 80,1,90
Cluster 1: 100,2,100
Cluster 2: 300,5,200
Missing values globally replaced with mean/mode
Status
OK
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Recharge
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
23:18:14 - SimpleKMeans
Clusterer output
Cluster 1: 100,2,100
Cluster 2: 300,5,200
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1 2
(5.0) (1.0) (1.0) (3.0)
CallDuration 230 80 100 323.3333
DataUsage 3.9 1 2 5.5
Recharge 164 90 100 210
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1 (20%)
1 1 (20%)
2 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Weka Clusterer Visualize: 23:18:14 - SimpleKMeans (telecom)
X: Instance_number (Num)
Y: CallDuration (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: telecom_clustered
350
215
80
0
2
4
X
Y
Class colour
cluster0 cluster1 cluster2
Status
OK
Log
x 0

Analysis
With **K = 2**, users can be grouped approximately as:

- **High-usage/high-value cluster:** U1, U3, U5
- **Low-usage/low-value cluster:** U2, U4

The low-usage group may represent customers with lower engagement and potentially higher churn risk.
Industry Application
The telecom company can provide personalized recharge plans, discounts, additional data benefits, and retention offers to low-engagement customers.
High-value customers can receive premium plans and loyalty benefits.
Conclusion
Clustering helps telecom companies understand customer usage patterns and design targeted retention strategies. For accurate churn prediction, clustering should later be combined with historical churn data.

Q4 . Dataset: E-commerce Behavior

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:
Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

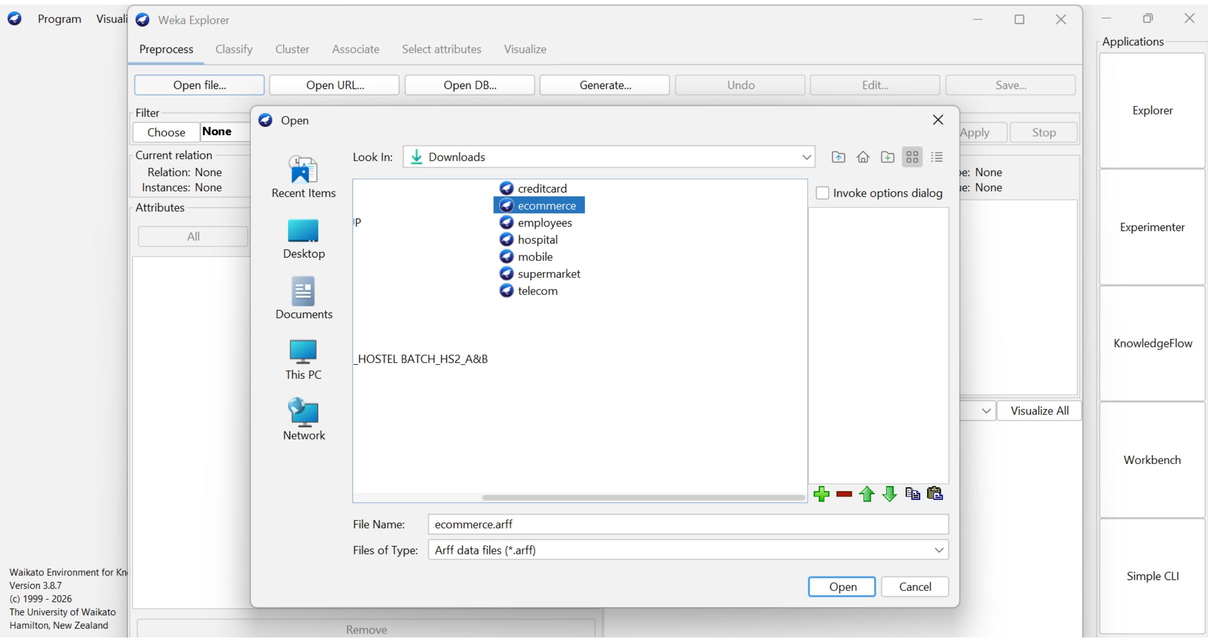
ANSWER: E-Commerce Behavior

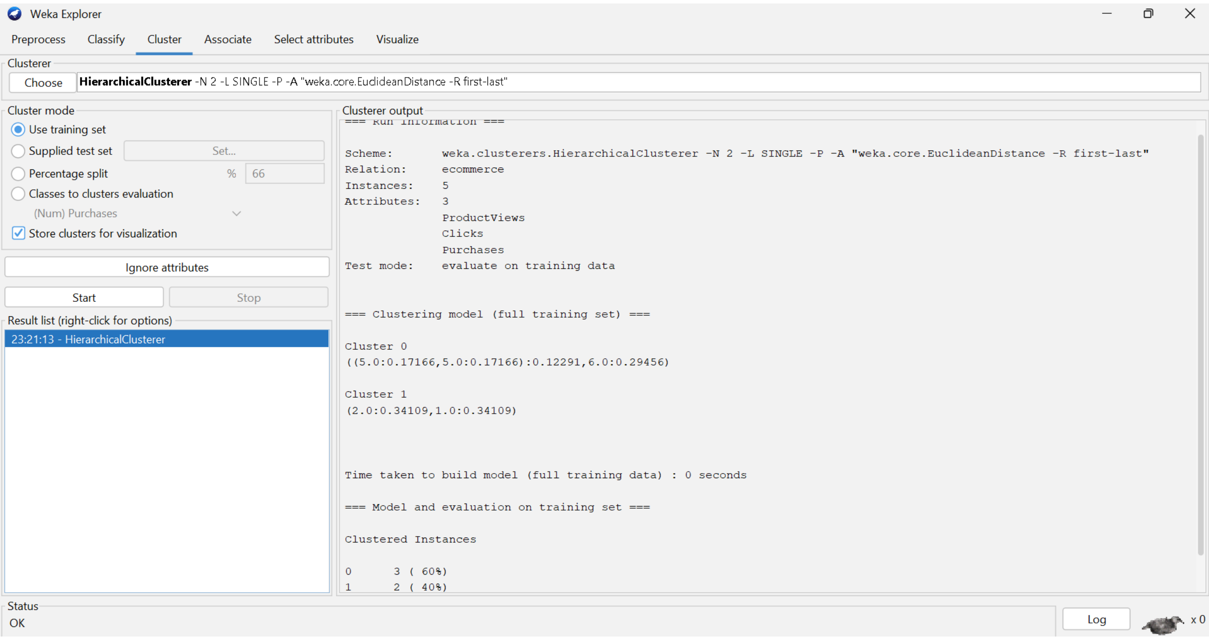
Clustering Technique: Hierarchical Clustering
Hierarchical clustering can group users according to their product views, clicks, and purchases. It is useful for discovering groups of users with similar browsing and purchasing behavior.
Input

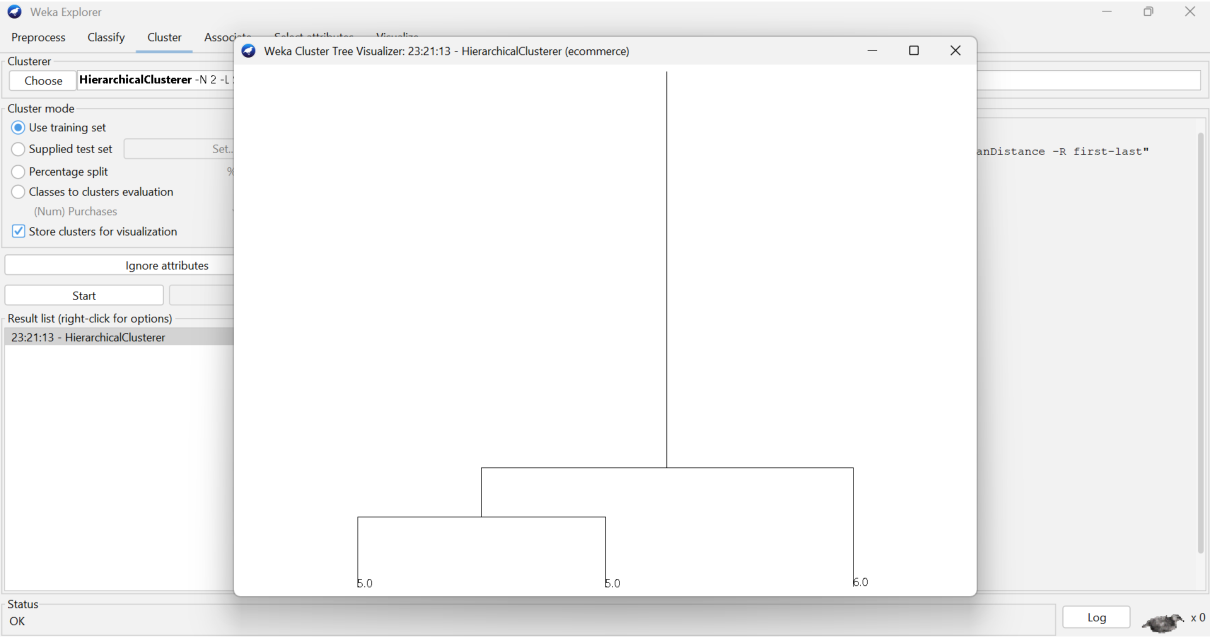
User	Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1

User	Views	Clicks	Purchases
U5	55	22	5

OUTPUT:







Analysis

Users U1, U3, and U5 show high engagement with many views, clicks, and purchases. Users U2 and U4 show relatively lower engagement.

Therefore, the users can be grouped into:

- **High-engagement customers:** U1, U3, U5
- **Low-engagement customers:** U2, U4

Industry Application

The recommendation system can use these clusters to provide personalized product recommendations.

High-engagement users can receive related products, premium products, and cross-selling recommendations. Low-engagement users can receive popular products, discounts, and promotional recommendations.

Conclusion

Hierarchical clustering helps identify behavioral customer groups, allowing e-commerce platforms to personalize recommendations and improve customer engagement.

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

ANSWER: Traffic Data

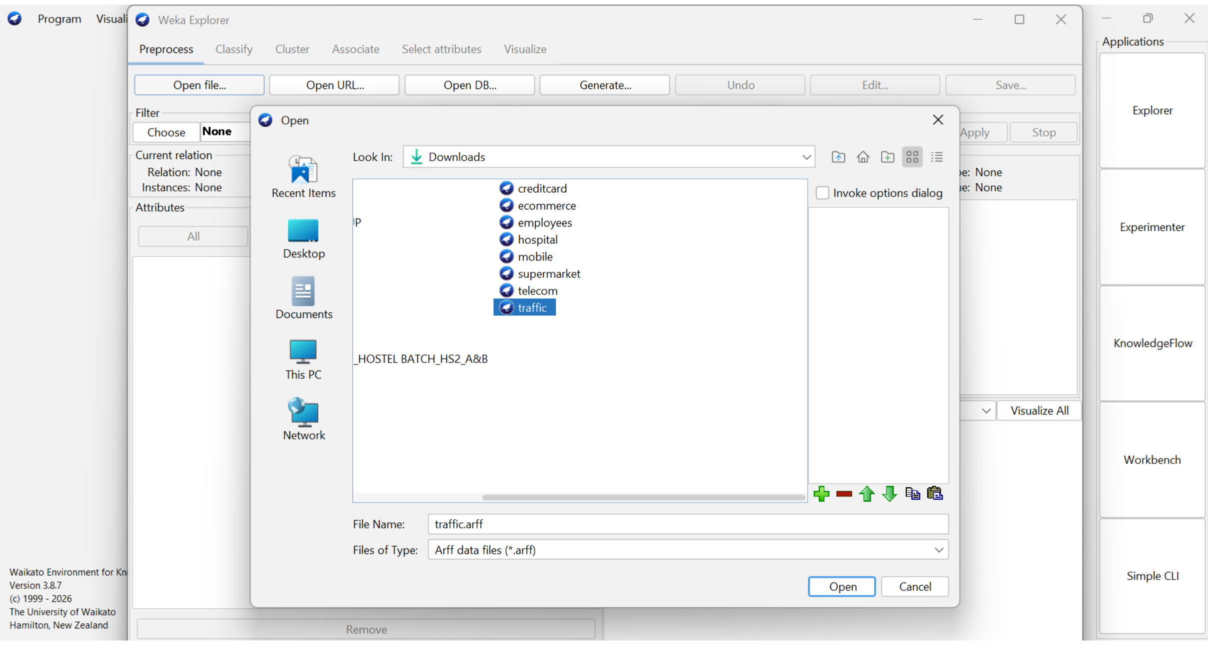
Clustering Technique: K-Means Clustering

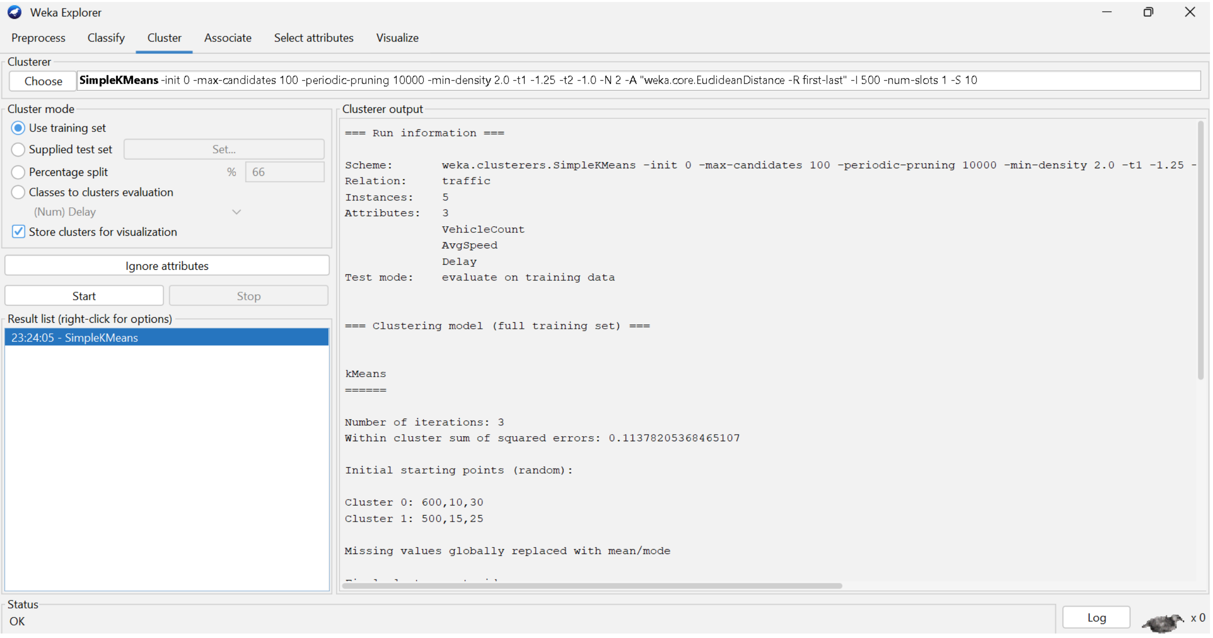
K-Means can group traffic locations using **vehicle count, average speed, and delay**.

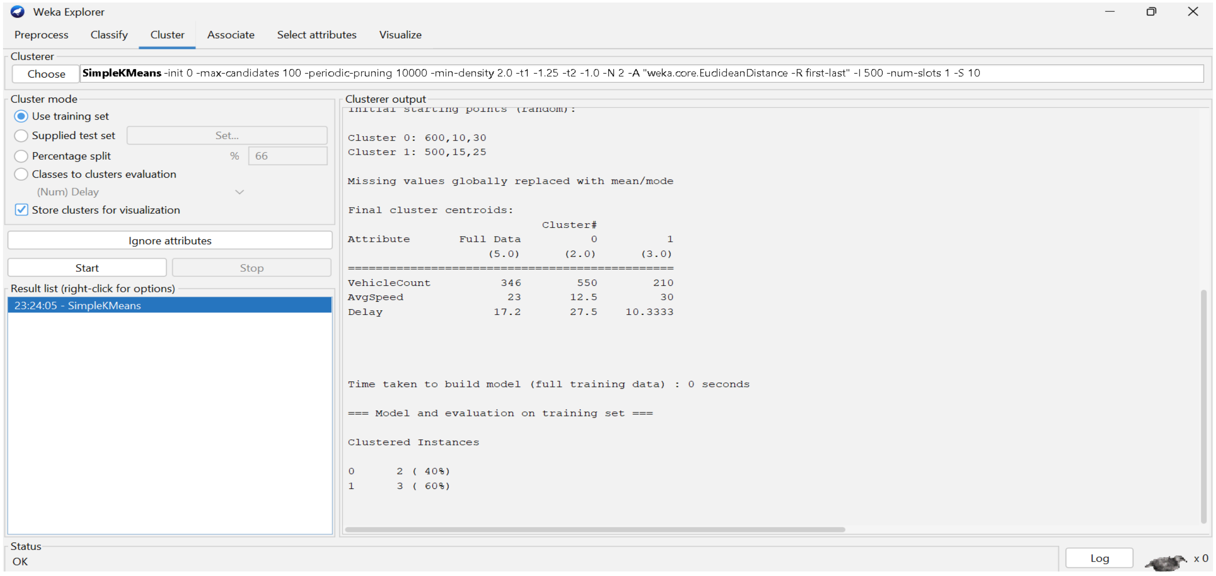
Input

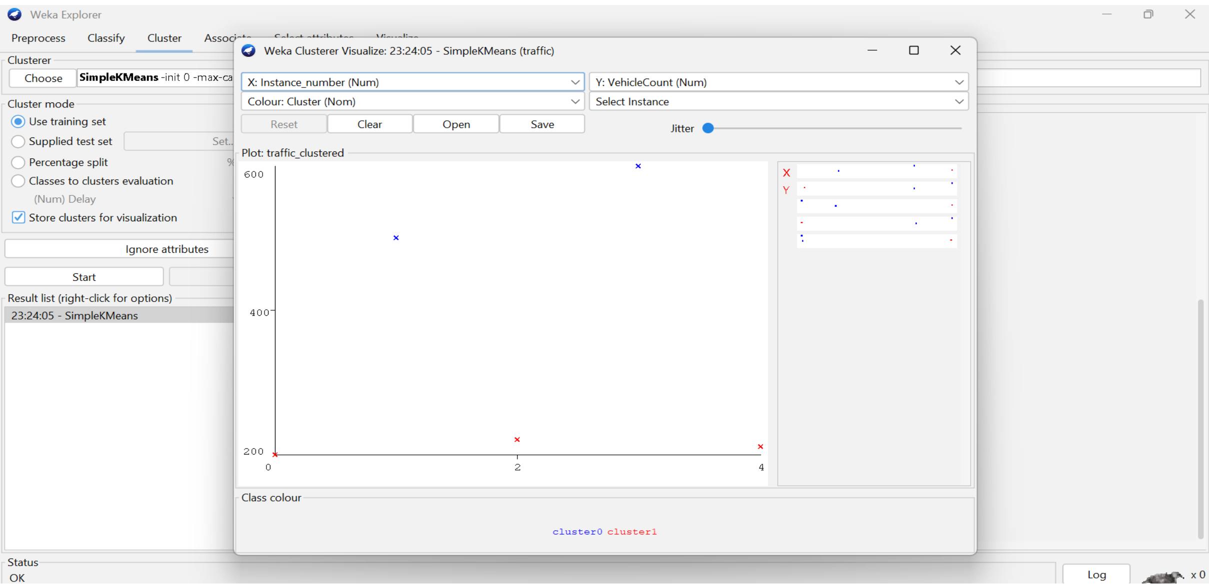
Location	Vehicles	Avg Speed	Delay
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

OUTPUT:




Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split % 66
Classes to clusters evaluation
(Num) Delay
Store clusters for visualization
Ignore attributes
Start Stop
Result list (right-click for options)
23:24:05 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation: traffic
Instances: 5
Attributes: 3
VehicleCount
AvgSpeed
Delay
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 3
Within cluster sum of squared errors: 0.11378205368465107
Initial starting points (random):
Cluster 0: 600,10,30
Cluster 1: 500,15,25
Missing values globally replaced with mean/mode
Status
OK
Log
x 0


Weka Explorer
Clusterer output
Cluster 0: 600,10,30
Cluster 1: 500,15,25
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
VehicleCount 346 550 210
AvgSpeed 23 12.5 30
Delay 17.2 27.5 10.3333
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK


Weka Clusterer Visualize: 23:24:05 - SimpleKMeans (traffic)
X: Instance_number (Num)
Y: VehicleCount (Num)
Colour: Cluster (Nom)
Select Instance
Reset Clear Open Save
Jitter
Plot: traffic_clustered
600
400
200
0
2
4
Class colour
cluster0 cluster1

Analysis

With **K = 2**, the locations can be grouped approximately as:

- **Low-congestion cluster:** L1, L3, L5
- **High-congestion cluster:** L2, L4

The high-congestion locations have high vehicle counts, low average speeds, and high delays.

Industry Application

Traffic authorities can use these clusters to:

- Adjust traffic signals
- Deploy traffic personnel
- Provide alternate routes
- Manage traffic flow
- Send congestion alerts

Conclusion

K-Means clustering helps identify traffic congestion zones by grouping locations with similar traffic conditions. This supports data-driven traffic management and better allocation of traffic-control resources.